

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
```

```
import os
print(os.listdir())
```

```
['.config', 'SampleSuperstore.csv', 'sample_data']
```

```
data = pd.read_csv('SampleSuperstore.csv', encoding='latin1')
data.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale

5 rows × 21 columns

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5 rows × 21 columns

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Row ID              9994 non-null  int64
1   Order ID            9994 non-null  object
2   Order Date          9994 non-null  object
3   Ship Date           9994 non-null  object
4   Ship Mode           9994 non-null  object
5   Customer ID         9994 non-null  object
6   Customer Name       9994 non-null  object
7   Segment             9994 non-null  object
8   Country             9994 non-null  object
9   City               9994 non-null  object
10  State              9994 non-null  object
11  Postal Code         9994 non-null  int64
12  Region             9994 non-null  object
13  Product ID          9994 non-null  object
14  Category            9994 non-null  object
15  Sub-Category        9994 non-null  object
16  Product Name        9994 non-null  object
17  Sales               9994 non-null  float64
18  Quantity            9994 non-null  int64
19  Discount            9994 non-null  float64
20  Profit              9994 non-null  float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```

```
data.isnull().sum()
```

	0
Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0

dtype: int64

```
data = data.dropna()
```

```
X = data[['Profit', 'Quantity']]
y = data['Sales']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

▼ LinearRegression ⓘ ?

```
LinearRegression()
```

```
predictions = model.predict(X_test)
```

```
comparison = pd.DataFrame({
    'Actual': y_test,
    'Predicted': predictions
})
```

	Actual	Predicted
3125	563.808	220.775519
1441	36.672	122.184450
4510	37.300	130.820895
39	212.058	123.813718
4509	171.288	137.036209

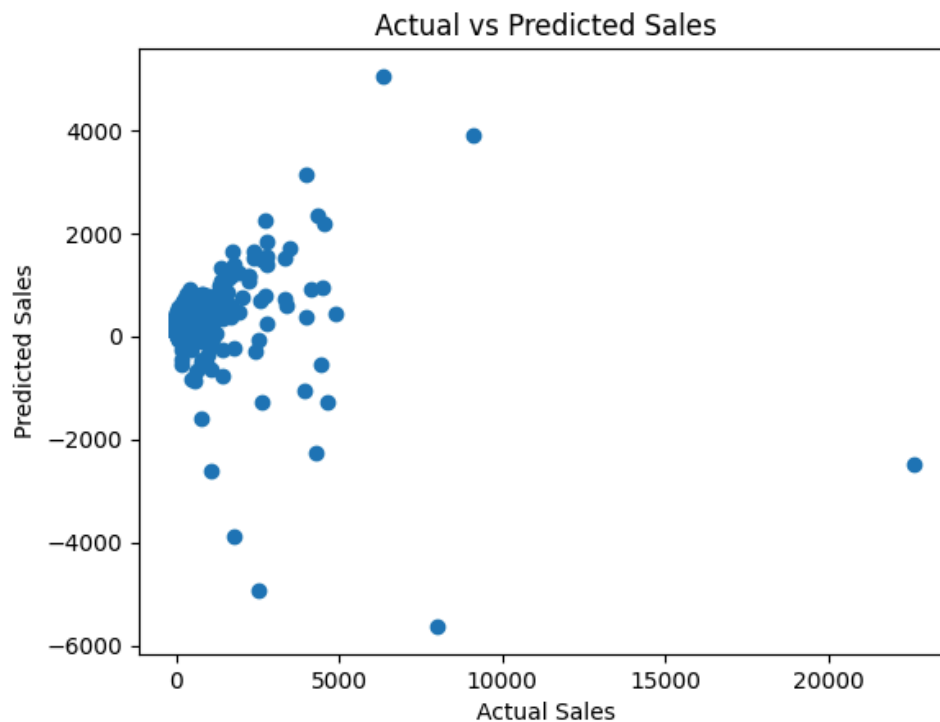
Next steps: [Generate code with comparison](#) [New interactive sheet](#)

```
mae = mean_absolute_error(y_test, predictions)
r2 = r2_score(y_test, predictions)
```

```
print("Mean Absolute Error:", mae)
print("R2 Score:", r2)
```

Mean Absolute Error: 249.29491447156843
R2 Score: -0.15358184652805051

```
plt.scatter(y_test, predictions)
plt.xlabel('Actual Sales')
plt.ylabel('Predicted Sales')
plt.title('Actual vs Predicted Sales')
plt.show()
```



```
future_prediction = model.predict([[200, 5]])
print("Predicted Sales:", future_prediction)
```

Predicted Sales: [533.82639812]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not
warnings.warn(

